Complexity analysis document

Provide a short report discussing:

- Reservation insertion complexity
- Reservation removal complexity
- Waiting-list processing complexity
- Undo cancellation complexity

Include Big-O notation and brief explanations.

1. Reservation insertion complexity
   Time Complexity: O(1)
   Space Complexity: O(1)
   Because a new reservation is inserted at the beginning (head) of the linked list. The insertion only takes constant time and space in memory.

2. Reservation removal complexity
   Time Complexity: O(n)
   Space Complexity: O(1)
   Because traverses all element to remove the desired reservation until it is found. In worst Case scenario, the element might be the last node of linked list. For space, to remove itself it does not require any other space.

3. Waiting-list processing complexity
   Time Complexity: O(1)
   Space Complexity: O(n)
   Because waiting list is implemented as a queue. Adding the element at end or removing element from beginning takes constant time as it is maintained using pointer. It requires O(n) space.

4. Undo cancellation complexity
   Time Complexity: O(1)
   Space Complexity: O(1)
   Because cancellation history is stores as stack. Performing Undo operation in stack takes constant time (removing element from top). And it takes constant space.

Reported by Om Mahato.